QUIZSPIN

1. Game Concept Document

Part	QuizSpin Answer
Title and one-line pitch	QuizSpin — A web-based educational spin-the-wheel game where players spin a wheel and answer interactive questions to earn points.
Genre and platform	Educational Quiz / Casual Game; Web Browser
Target players	High school students, college students, teachers/instructors, and learners who want an engagin g quiz experience.
Core loop	Spin the wheel → Get a question → Choose an answer → Earn points → Continue playing.
Mechanics and controls	Click Spin; receive a randomized question choose an answer; server checks the answer; score updates.
Win/lose rules	Correct answers earn points. Wrong answers earn no points. At game end, the highest score wins.
Must Have	Spin wheel, questions, multiple-choice answers, answer checking, score tracking, join code, leaderboard.
Should Have	Categories, timer, randomized questions, nickname.
Could Have	Sound effects, animations, themes, difficulty levels.
Won't Have	Real-money rewards, complex character customization, offline multiplayer.
Backend + Hub fit	Host login, question sets, sessions, 6-digit join codes, player answers, validation, scores, leaderboard; game-info and health endpoints; no correct answer sent to players.

2. User Story Backlog with MoSCoW

ID	User Story	Priority
US-01	As a host, I want to log in so that I can manage a QuizSpin game.	Must
US-02	As a host, I want to choose a question set so that I can prepare the game.	Must
US-03	As a host, I want to start a game so that players can join.	Must
US-04	As a player, I want to join using a 6-digit code and nickname so that I can play without an account.	Must
US-05	As a player, I want to spin the wheel so that I receive a question.	Must
US-06	As a player, I want to submit an answer so that the system can check it.	Must
US-07	As a player, I want my score updated after an answer so that I	Must

	know my progress.	
US-08	As a player, I want to view the leaderboard so that I know my rank.	Must
US-09	As a host, I want different question categories so that games can cover different topics.	Should
US-10	As a player, I want a timer so that the game feels more challenging.	Should
US-11	As a player, I want sound effects and animations so that the game is more engaging.	Could
US-12	As a player, I want difficulty levels so that I can choose an appropriate challenge.	Could
US-13	As a player, I want offline multiplayer so that I can play without a network connection.	Won't

3. Use Case Diagram + 3 Narratives

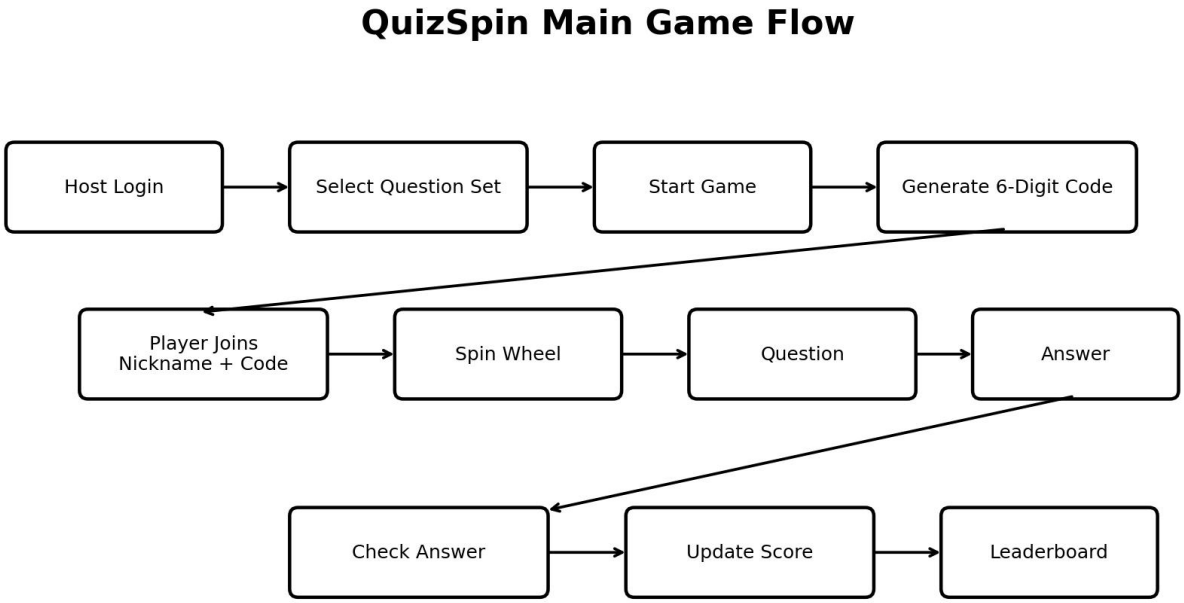
Use Case	Actor	Brief Narrative
UC-01 Login	Host	Host enters credentials. System validates them and allows access when valid.
UC-02 Start/Join Game	Host / Player	Host selects a question set and starts a session. System generates a 6-digit code. Player enters the code and nickname to join.
UC-03 Submit Answer	Player	Player submits a selected choice. System validates the input, checks the answer, updates the score, and returns the result without exposing the correct answer.

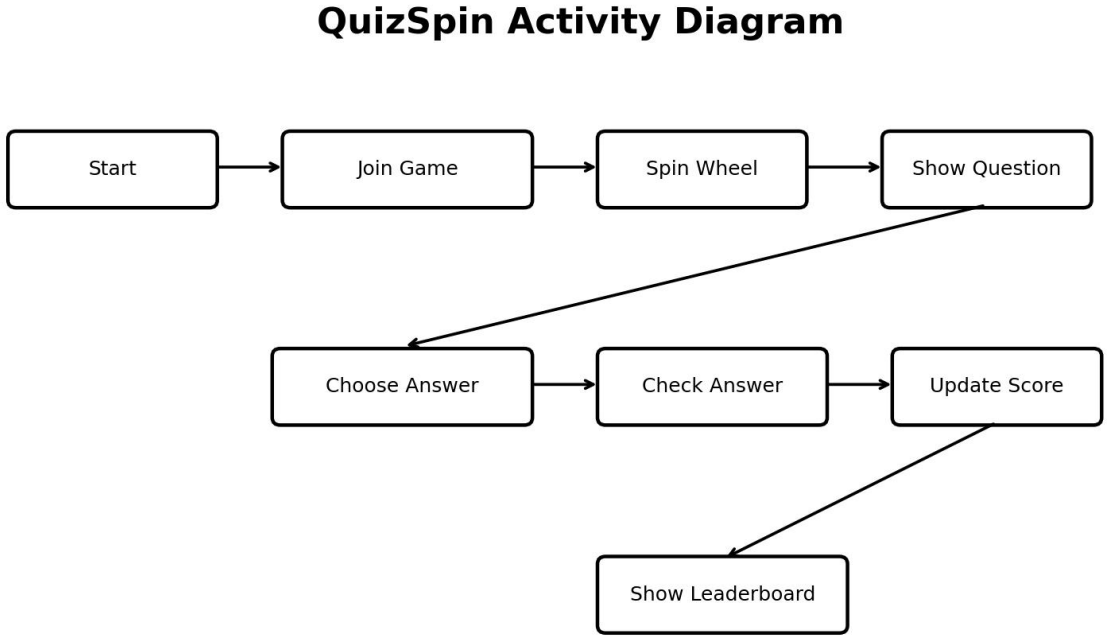
4. Class Diagram

Class	Key Attributes	Main Responsibilities
Host	hostId, username, password	Authenticate and manage question sets/sessions.
QuestionSet	questionSetId, title, description	Store a collection of questions.
Question	questionId, questionSetId, text, choices, correctChoice	Provide quiz content; correct choice remains server-side.
GameSession	sessionId, code, questionSetId, status	Manage an active QuizSpin game.
Player	playerId, sessionId, nickname, score	Represent a player in a session and track score.
Answer	answerId, playerId, questionId, choiceIndex, isCorrect	Record a player's submitted choice and result.

Leaderboard	sessionId, playerId, rank, score	Provide player rankings for a session.

5. Sequence + Activity Diagrams





6. ERD + Schema + Data Dictionary

Table	Primary Key	Important Foreign Keys	Purpose
hosts	host_id	—	Stores host login accounts.
question_sets	question_set_id	host_id	Stores quiz/question-set information.
questions	question_id	question_set_id	Stores questions and server-side answer information.
game_sessions	session_id	question_set_id	Stores active/completed game sessions and join codes.
players	player_id	session_id	Stores player nicknames and scores.
answers	answer_id	player_id, question_id	Stores submitted choices and checking results.

7. API Endpoint Table

Verb	Endpoint	What it does	Token	Success	Errors
GET	/api/v1/game-info	Returns QuizSpin name, description and details.	No	200	500
GET	/api/v1/health	Checks API/database health.	No	200	503
POST	/api/v1/auth/login	Host logs in.	No	200	400, 401
POST	/api/v1/question-sets	Create a question set.	Host	201	400, 401
GET	/api/v1/question-sets	List question sets.	Host	200	401
POST	/api/v1/sessions	Start a QuizSpin session.	Host	201	400, 401, 404
POST	/api/v1/sessions/{code}/players	Player joins with nickname.	No	201	400, 404, 409
POST	/api/v1/sessions/{code}/answers	Submit a player choice.	Player	201	400, 401, 409
GET	/api/v1/sessions/{code}/leaderboard	View rankings.	No	200	404

8. Architecture, Component, Deployment Diagrams

Component	Responsibility
API Router	Receives HTTP requests and maps URLs to controllers.
Auth	Handles host authentication.
Question Sets	Creates and retrieves question sets.

Sessions & Players	Creates sessions, join codes, and players.
Answers & Leaderboard	Checks answers, updates scores, and returns rankings.
Repository	Contains database/SQL operations.
Database	Stores hosts, questions, sessions, players, and answers.